

TEAM_10

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Mini Sentries Hackathon

Day 1 — Foundation & Architecture

Environment Summary

The Day 1 environment consists of three virtual machines connected on a shared host-only/bridged network segment, with Suricata IDS configured to monitor traffic between them.

Kali Linux (Attacker VM): 172.17.216.154

Metasploitable2 (Victim / Target VM): 172.17.216.69

Ubuntu VM (Security Stack / Suricata host): 172.17.216.84

Note: Suricata, deployed on the Ubuntu VM, monitors the segment connecting the Kali attacker machine and the Metasploitable2 victim machine, in line with the architecture diagram below.

1. Architecture Diagram

The finalised Day 1 architecture diagram, showing all four mandatory layers (Perimeter & Detection, SIEM & Visibility, Identity & Access, Automated Response) and the data flow between them.

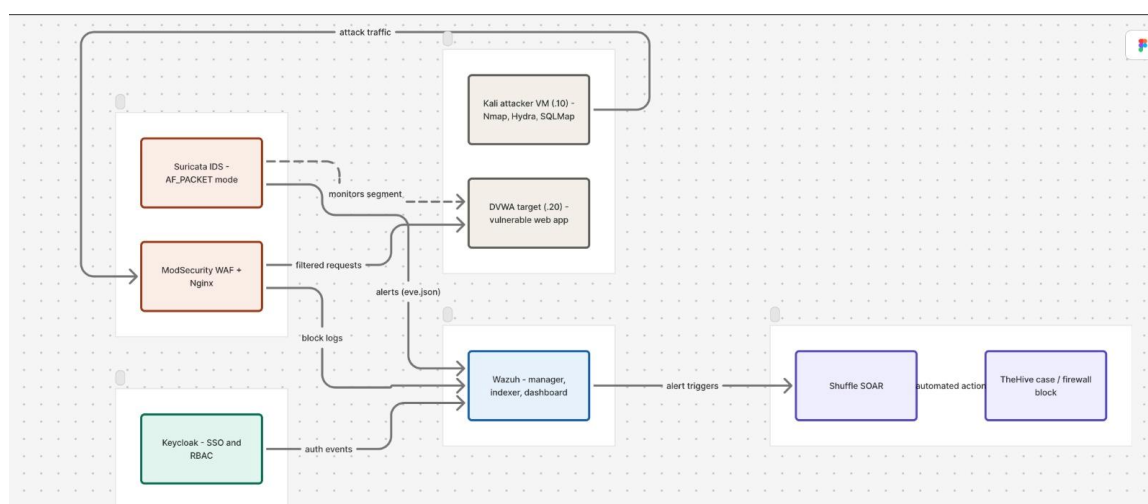


Figure 1: Mini Sentries architecture diagram — Kali attacker VM and DVWA target sit behind Suricata IDS and ModSecurity WAF; alerts and logs flow into Wazuh, which triggers Shuffle SOAR and TheHive for automated response. Keycloak provides identity/auth events into the same SIEM pipeline.

2. Virtual Machine Setup & Network Connectivity

2.1 Kali Linux Attacker VM

Kali Linux VM configured with IP 172.17.216.154 on the shared network segment. Connectivity to the victim machine (172.17.216.69) was verified using ping with a 0% packet loss result.

```
(root@kali)-[~]
# ip addr show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5c:e0:2f brd ff:ff:ff:ff:ff:ff
    inet 172.17.216.154/24 brd 172.17.216.255 scope global dynamic noprefixroute eth0
        valid_lft 3259sec preferred_lft 3259sec
    inet6 2409:40f4:4107:3a8b:e9d3:7b70:c0f7:2565/64 scope global temporary dynamic
        valid_lft 6863sec preferred_lft 6863sec
    inet6 2409:40f4:4107:3a8b:a00:27ff:fe5c:e02f/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 6863sec preferred_lft 6863sec
    inet6 fe80::a00:27ff:fe5c:e02f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(root@kali)-[~]
# ping 172.17.216.69
PING 172.17.216.69 (172.17.216.69) 56(84) bytes of data.
64 bytes from 172.17.216.69: icmp_seq=1 ttl=64 time=70.1 ms
64 bytes from 172.17.216.69: icmp_seq=2 ttl=64 time=84.7 ms
64 bytes from 172.17.216.69: icmp_seq=3 ttl=64 time=106 ms
64 bytes from 172.17.216.69: icmp_seq=4 ttl=64 time=93.8 ms
64 bytes from 172.17.216.69: icmp_seq=5 ttl=64 time=118 ms
64 bytes from 172.17.216.69: icmp_seq=6 ttl=64 time=45.6 ms
64 bytes from 172.17.216.69: icmp_seq=7 ttl=64 time=11.1 ms
64 bytes from 172.17.216.69: icmp_seq=8 ttl=64 time=8.59 ms
^C
--- 172.17.216.69 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7195ms
rtt min/avg/max/mdev = 8.585/67.229/118.003/38.973 ms

(root@kali)-[~]
```

Figure 2: Kali VM (eth0 = 172.17.216.154) successfully pinging the Metasploitable2 victim VM at 172.17.216.69 — 8 packets transmitted, 0% packet loss, confirming network connectivity.

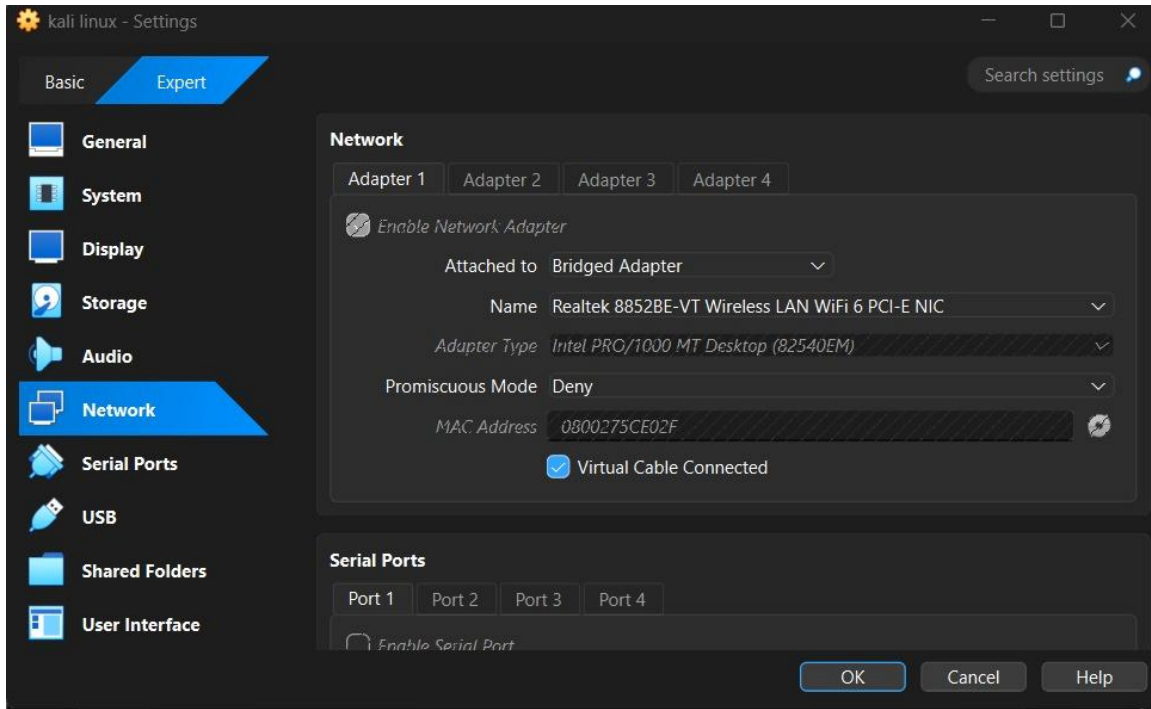


Figure 3: VirtualBox network settings for the Kali Linux VM — Adapter 1 set to Bridged Adapter mode using the host's wireless NIC, with Promiscuous Mode set to Deny (per SRM network restrictions).

2.2 Metasploitable2 Target VM

The Metasploitable2 target VM was used to verify reachability of the other VMs on the segment and to confirm open/vulnerable services for later attack simulations.

```

8 packets transmitted, 0 received, +6 errors, 100% packet loss, time 7042ms
, pipe 3
msfadmin@metasploitable:~$ ping 172.17.216.84
PING 172.17.216.84 (172.17.216.84) 56(84) bytes of data:
64 bytes from 172.17.216.84: icmp_seq=1 ttl=64 time=84.1 ms
64 bytes from 172.17.216.84: icmp_seq=2 ttl=64 time=26.4 ms
64 bytes from 172.17.216.84: icmp_seq=3 ttl=64 time=103 ms

--- 172.17.216.84 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2012ms
rtt min/avg/max/mdev = 26.477/71.367/103.486/32.711 ms
msfadmin@metasploitable:~$ ping 172.17.216.154
PING 172.17.216.154 (172.17.216.154) 56(84) bytes of data:
64 bytes from 172.17.216.154: icmp_seq=1 ttl=64 time=59.8 ms
64 bytes from 172.17.216.154: icmp_seq=2 ttl=64 time=23.2 ms
64 bytes from 172.17.216.154: icmp_seq=3 ttl=64 time=16.1 ms
64 bytes from 172.17.216.154: icmp_seq=4 ttl=64 time=32.6 ms
64 bytes from 172.17.216.154: icmp_seq=5 ttl=64 time=12.4 ms
64 bytes from 172.17.216.154: icmp_seq=6 ttl=64 time=12.1 ms
64 bytes from 172.17.216.154: icmp_seq=7 ttl=64 time=10.6 ms

--- 172.17.216.154 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6021ms
rtt min/avg/max/mdev = 10.674/23.876/59.876/16.364 ms
msfadmin@metasploitable:~$

```

Figure 4: Metasploitable2 VM pinging the Ubuntu security-stack VM (172.17.216.84) and the Kali attacker VM (172.17.216.154) — both reachable with 0% packet loss, confirming all three VMs are on the same network segment.

```

udp      0      0 0.0.0.0:44684      0.0.0.0:*
4956/rpc.mountd
udp      0      0 172.17.216.69:53   0.0.0.0:*
4642/named
udp      0      0 127.0.0.1:53        0.0.0.0:*
4642/named
udp      0      0 0.0.0.0:68          0.0.0.0:*
5271/dhclient3
udp      0      0 0.0.0.0:69          0.0.0.0:*
5047/xinetd
udp      0      0 0.0.0.0:60888       0.0.0.0:*
-
udp      0      0 0.0.0.0:50396       0.0.0.0:*
4642/named
udp      0      0 0.0.0.0:111         0.0.0.0:*
4260/portmap
udp      0      0 0.0.0.0:36848       0.0.0.0:*
4276/rpc.statd
udp      0      0 0.0.0.0:636         0.0.0.0:*
4276/rpc.statd
udp6     0      0 :::50867            :::*
4642/named
udp6     0      0 :::53               :::*
4642/named
msfadmin@metasploitable:~$

```

Figure 5: netstat output on the Metasploitable2 VM listing active UDP listeners (rpc.mountd, named/DNS, dhclient, xinetd, portmap, rpc.statd) — used to enumerate baseline services running on the target before attack simulation.

2.3 Reconnaissance Scan Against the Target

An Nmap service-version scan was run from the Kali VM against the Metasploitable2 target (172.17.216.69) to enumerate open ports and running services, providing the baseline attack-surface map referenced by the architecture diagram.

```

(root@kali)-[~]
# sudo nmap -Pn -sT -sV 172.17.216.69
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-19 15:46 IST
Nmap scan report for 172.17.216.69
Host is up (0.016s latency).
Not shown: 978 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.77 seconds

```

Figure 6: Nmap scan (-Pn -sT -sV) from Kali against 172.17.216.69, revealing 23 open ports including FTP, SSH, Telnet, SMTP, Samba, MySQL, PostgreSQL, VNC and other classic Metasploitable2 vulnerable services.

2.4 Ubuntu Security-Stack VM

The Ubuntu VM hosts the Suricata IDS and Wazuh agent. Its connectivity to both the victim and attacker VMs was verified, and its network adapter configuration was iterated to satisfy SRM's host-only/bridged network constraints.

```
vbuser@Ubuntu:~$ ping 172.17.216.69
PING 172.17.216.69 (172.17.216.69) 56(84) bytes of data.
64 bytes from 172.17.216.69: icmp_seq=1 ttl=64 time=312 ms
64 bytes from 172.17.216.69: icmp_seq=2 ttl=64 time=111 ms
64 bytes from 172.17.216.69: icmp_seq=3 ttl=64 time=286 ms
64 bytes from 172.17.216.69: icmp_seq=4 ttl=64 time=139 ms
^C
--- 172.17.216.69 ping statistics ---
5 packets transmitted, 4 received, 20% packet loss, time 4036ms
rtt min/avg/max/mdev = 110.606/211.988/312.423/88.392 ms
vbuser@Ubuntu:~$ ping 172.17.216.154
PING 172.17.216.154 (172.17.216.154) 56(84) bytes of data.
64 bytes from 172.17.216.154: icmp_seq=1 ttl=64 time=84.5 ms
64 bytes from 172.17.216.154: icmp_seq=2 ttl=64 time=16.0 ms
64 bytes from 172.17.216.154: icmp_seq=3 ttl=64 time=16.9 ms
64 bytes from 172.17.216.154: icmp_seq=4 ttl=64 time=63.1 ms
^C
--- 172.17.216.154 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3083ms
rtt min/avg/max/mdev = 16.044/45.147/84.543/29.665 ms
vbuser@Ubuntu:~$
```

Figure 7: Ubuntu VM pinging the Metasploitable2 victim (172.17.216.69) and Kali attacker (172.17.216.154) VMs — both reachable, confirming the security-stack VM sits on the same monitored segment.

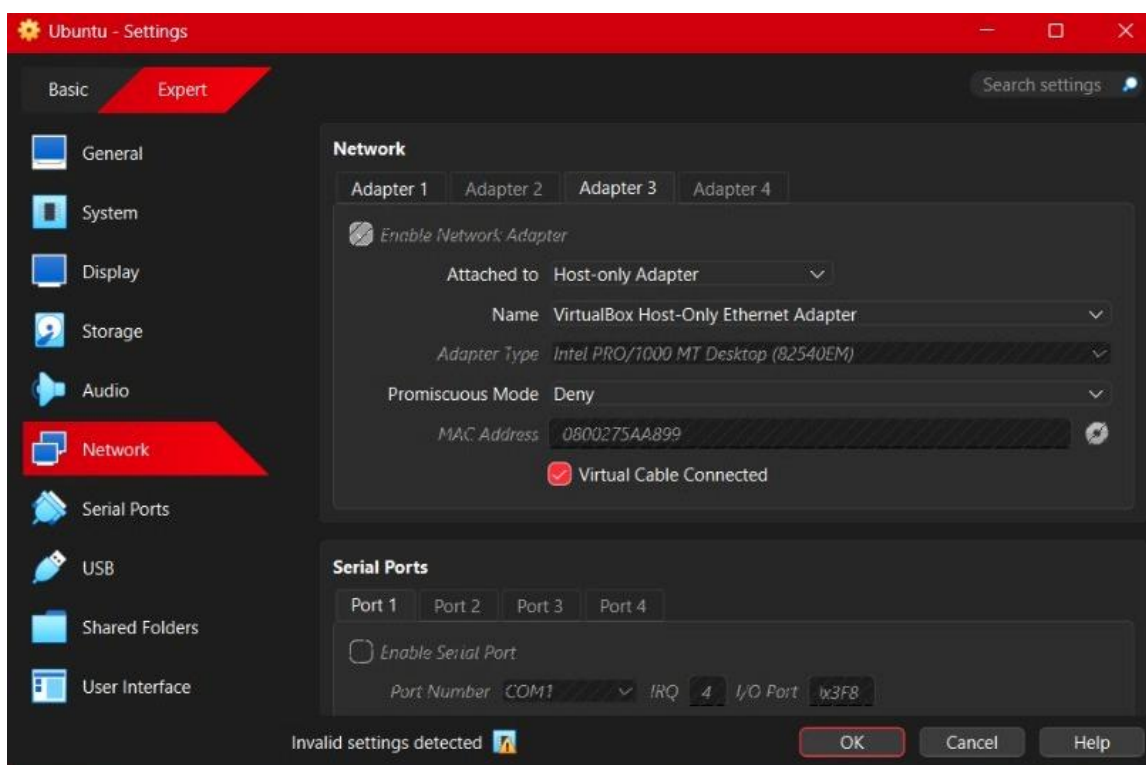


Figure 8: Initial VirtualBox network configuration attempt for the Ubuntu VM — Adapter 3 set to Host-only Adapter. The settings dialog flags "Invalid settings detected," prompting a reconfiguration.

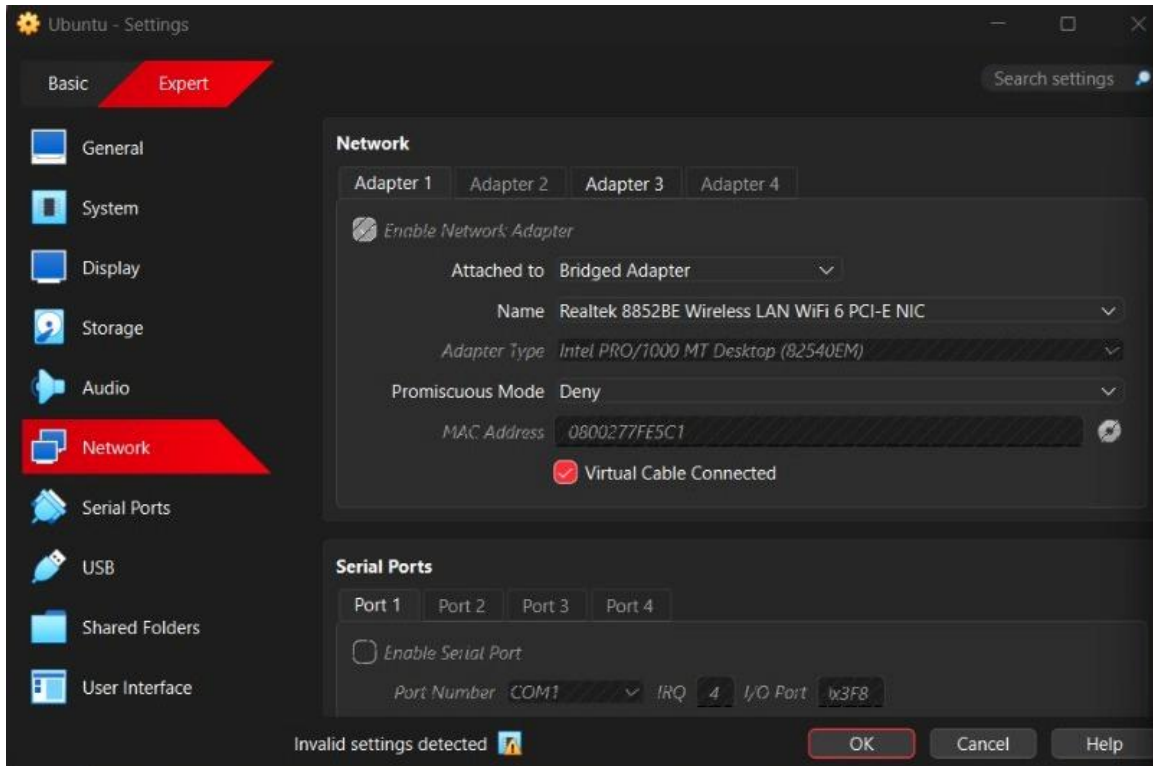


Figure 9: Corrected VirtualBox network configuration for the Ubuntu VM — Adapter 3 switched to Bridged Adapter using the host's wireless NIC to resolve the invalid-settings warning and align with the other VMs.

3. IDS Rule Configuration

Custom Suricata detection rules were written to cover the four key attack stages the team intends to simulate across the hackathon: reconnaissance, SSH brute force, SQL injection, and reverse shell activity.

```
vbuser@Ubuntu:~$ cat /etc/suricata/rules/suricata.rules
# Rule 1 - Reconnaissance
alert tcp any any -> $HOME_NET any (
  msg:"Reconnaissance Activity Detected";
  flags:S;
  threshold:type both, track by_src, count 30, seconds 10;
  sid:1000001;
  rev:1;
)

# Rule 2 - SSH Brute Force
alert tcp any any -> $HOME_NET 22 (
  msg:"Possible SSH Brute Force";
  threshold:type both, track by_src, count 10, seconds 60;
  sid:1000002;
  rev:1;
)

# Rule 3 - SQL Injection
alert http any any -> $HOME_NET any (
  msg:"Possible SQL Injection";
  content:"union select"; nocase;
  sid:1000003;
  rev:1;
)

# Rule 4 - Reverse Shell Connection
alert tcp $HOME_NET any -> any 4444 (
  msg:"Possible Reverse Shell";
  sid:1000004;
  rev:1;
)
```

Figure 10: Contents of /etc/suricata/rules/suricata.rules on the Ubuntu VM — four custom rules defined for Reconnaissance Activity, SSH Brute Force, SQL Injection, and Reverse Shell Connection detection (sid 1000001–1000004).

4. Target VM Resource Configuration

The Metasploitable2 VM's virtual hardware was reviewed to ensure it has adequate resources to run reliably during attack simulations and live demos.

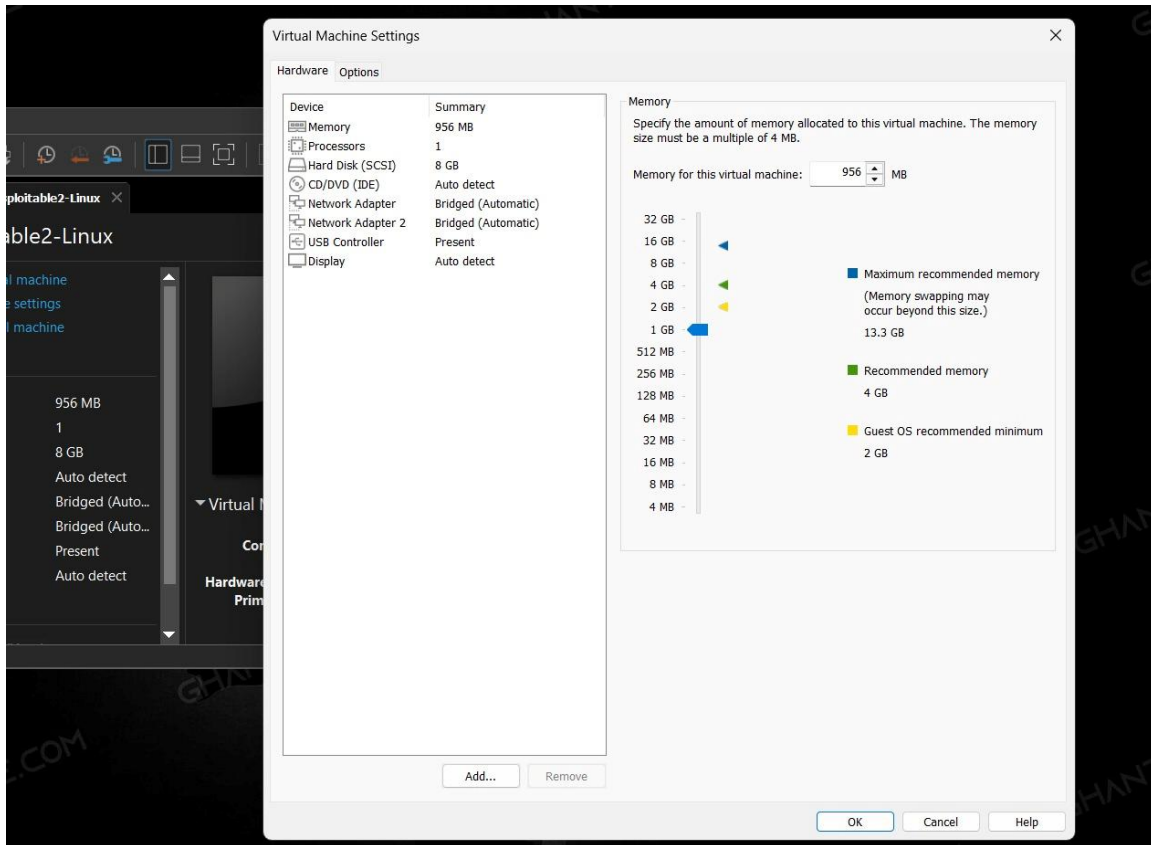


Figure 11: VirtualBox Virtual Machine Settings for Metasploitable2-Linux — 956 MB memory, 1 processor, 8 GB hard disk, two bridged network adapters, confirming the target VM's allocated resources.

5. Wazuh Agent Deployment

The Wazuh agent was installed and brought online on the Ubuntu VM to begin forwarding logs to the SIEM layer, in line with the SIEM & Visibility requirement of the architecture.

```

vbuser@Ubuntu:~$ sudo systemctl status wazuh-agent -l
● wazuh-agent.service - Wazuh agent
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-06-19 19:07:39 IST; 2min 41s ago
     Process: 7662 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0)
    Tasks: 34 (limit: 19089)
   Memory: 100.6M
      CPU: 15.739s
   CGroup: /system.slice/wazuh-agent.service
           └─7684 /var/ossec/bin/wazuh-execd
             └─7695 /var/ossec/bin/wazuh-agentd
               └─7709 /var/ossec/bin/wazuh-syscheckd
                 └─7722 /var/ossec/bin/wazuh-logcollector
                   └─7739 /var/ossec/bin/wazuh-modulesd

Jun 19 19:07:30 Ubuntu systemd[1]: Starting Wazuh agent...
Jun 19 19:07:30 Ubuntu env[7662]: Starting Wazuh v4.7.3...
Jun 19 19:07:32 Ubuntu env[7662]: Started wazuh-execd...
Jun 19 19:07:33 Ubuntu env[7662]: Started wazuh-agentd...
Jun 19 19:07:34 Ubuntu env[7662]: Started wazuh-syscheckd...
Jun 19 19:07:35 Ubuntu env[7662]: Started wazuh-logcollector...
Jun 19 19:07:37 Ubuntu env[7662]: Started wazuh-modulesd...
Jun 19 19:07:39 Ubuntu env[7662]: Completed.
Jun 19 19:07:39 Ubuntu systemd[1]: Started Wazuh agent.

vbuser@Ubuntu:~$ sudo tail -f /var/ossec/logs/ossec.log
2026/06/19 19:07:35 sca: INFO: Module started.
2026/06/19 19:07:35 sca: INFO: Loaded policy '/var/ossec/ruleset/sca/cis_ubuntu22-04.yml'
2026/06/19 19:07:35 wazuh-modulesd:control: INFO: Starting control thread.
2026/06/19 19:07:35 sca: INFO: Starting Security Configuration Assessment scan.
2026/06/19 19:07:35 wazuh-modulesd:syscollector: INFO: Module started.
2026/06/19 19:07:35 wazuh-modulesd:syscollector: INFO: Starting evaluation.
2026/06/19 19:07:36 sca: INFO: Starting evaluation of policy: '/var/ossec/ruleset/sca/cis_ubuntu22-04.yml'
2026/06/19 19:07:40 wazuh-modulesd:syscollector: INFO: Evaluation finished.
2026/06/19 19:09:46 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[172.31.188.215]:1515'
2026/06/19 19:09:51 wazuh-agentd: INFO: Requesting a key from server: 172.31.188.215

```

Figure 12: systemctl status confirming the wazuh-agent service is active (running) on the Ubuntu VM, with the OSSEC log tail showing successful startup of execd, agentd, syscheckd, logcollector and modulesd modules, and an in-progress request to enroll with the Wazuh manager.

6. Day 1 Brownie Point — "The Sleepwalker's Diary"

Decode: the riddle describes a logging system that records events but never correlates them across time — each shift/session is treated as a fresh start with no memory of prior entries. A real attacker exploiting this blind spot would spread an attack out over multiple days/sessions (a slow, low-and-slow campaign) so that no single alert ever looks suspicious in isolation, even though the full sequence — recon, then credential access, then discovery, then exploitation — is clearly a coordinated campaign when viewed together.

Implementation: a custom "Sleepwalker Attack Analyzer" script (sleepwalker.sh) was built to correlate alerts by source IP across multiple days, reconstructing the attack timeline per attacker and classifying campaigns (e.g. Low-and-Slow Attack) versus isolated, low-threat activity that lacks a full kill-chain pattern.

```
kali@kali: ~/Day1_Brownie
Session Actions Edit View Help
Log Out...

(kali@kali)~[~/Day1_Brownie]
$ ./sleepwalker.sh

=====
SLEEPWALKER ATTACK ANALYZER
=====

Attacker IP : 192.168.56.10

Timeline:

2026-06-19 → NMAP_SCAN
2026-06-21 → SSH_ATTEMPT
2026-06-24 → WEB_ENUMERATION
2026-06-27 → SQL_INJECTION

✓ Reconnaissance
✓ Credential Access
✓ Discovery
✓ Exploitation

ATTACK CAMPAIGN DETECTED
Campaign Type : Low-and-Slow Attack
Defender Blind Spot : Alerts treated independently
Prediction : Future compromise likely
Threat Level : CRITICAL

=====

Attacker IP : 192.168.56.20

Timeline:

2026-06-20 → NMAP_SCAN

✓ Reconnaissance

No Complete Campaign Detected
Threat Level : LOW
```

Figure 13: Output of `./sleepwalker.sh` — the Sleepwalker Attack Analyzer correlates multi-day activity from attacker IP 192.168.56.10 (Nmap scan -> SSH attempt -> web enumeration -> SQL injection across four separate days) into a single "Low-and-Slow Attack" campaign rated CRITICAL, versus attacker IP 192.168.56.20 which shows only an isolated recon scan rated LOW.